Skill Experiment 1: IAM User with Programmatic Access and AWS Billing Alerts

Aim

To create an IAM user that has programmatic access only, generate and download its Access Key ID and Secret Access Key, and set up AWS billing alerts so that I get an email when my Free Tier usage or spending goes over the limit.

Part A: Creating an IAM user with programmatic access only

Step 1: Sign in and open IAM

I signed in to the AWS Management Console with my root user. In the search bar I typed IAM and opened Identity and Access Management (IAM).

Step 2: Start creating the user

From the left menu I clicked Users and then clicked Create user.

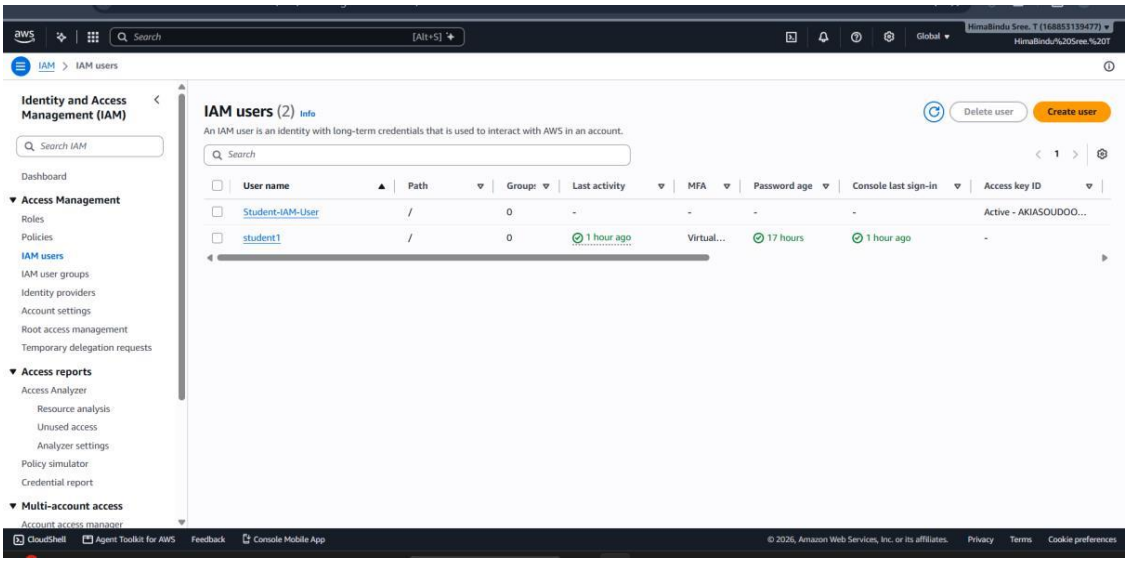


Figure 1: IAM users page with the Create user button

Step 3: Enter the username

I typed the username (Student-IAM-User). I did not tick the option for AWS Management Console access, because this user only needs programmatic access (through access keys). Then I clicked Next.

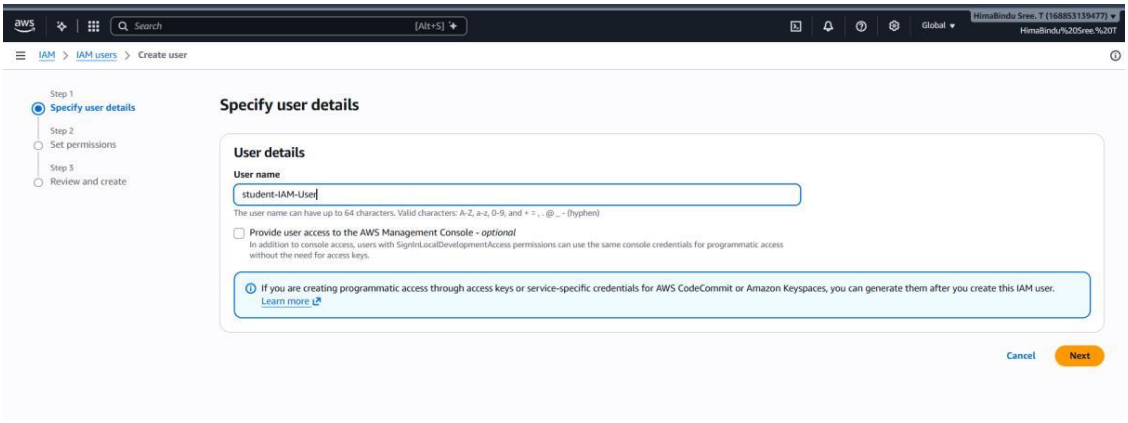


Figure 2: Specify user details page (console access not ticked)

Step 4: Attach the permissions

On Set permissions I selected Attach policies directly and ticked AdministratorAccess, since this lab needs administrative permissions. Then I clicked Next.

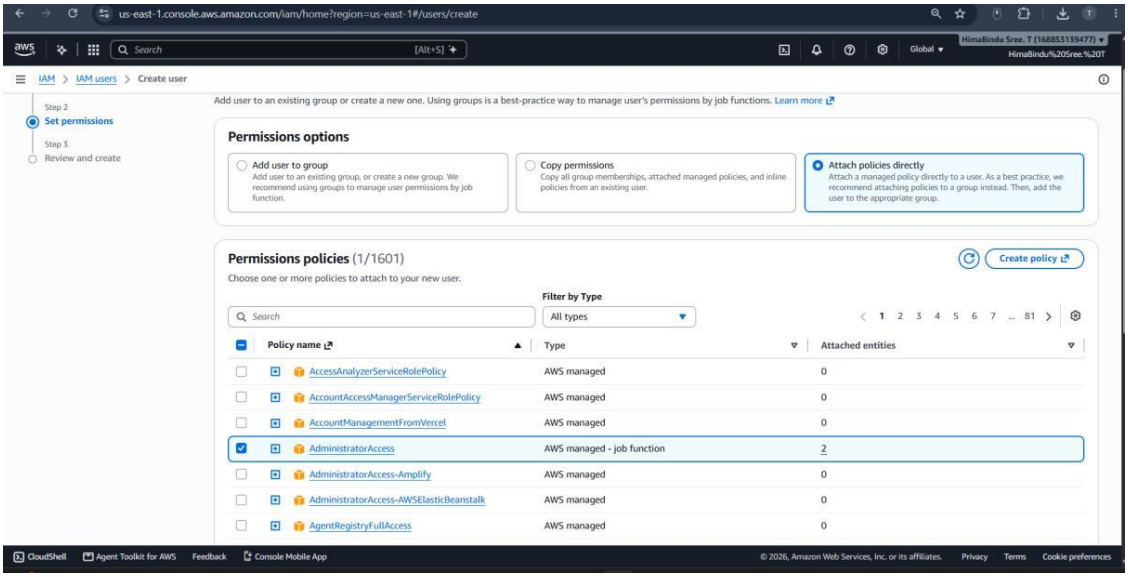


Figure 3: Set permissions page with AdministratorAccess selected

Step 5: Review and create

I checked the details on the review page and clicked Create user.

Step 6: Check the user list

AWS showed the message User created successfully. I went back to the Users list and confirmed that the new IAM user appears there. On the user's page, Console access shows as Disabled and the AdministratorAccess policy is attached.

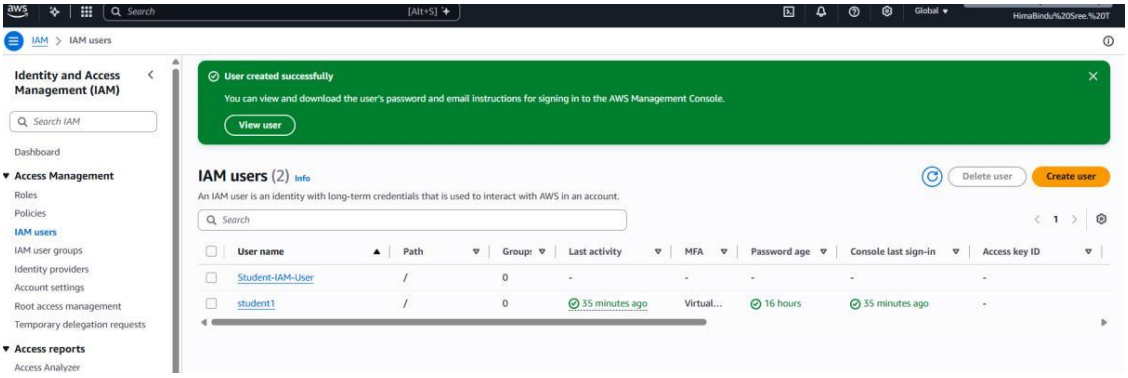


Figure 4: User created successfully – Users list

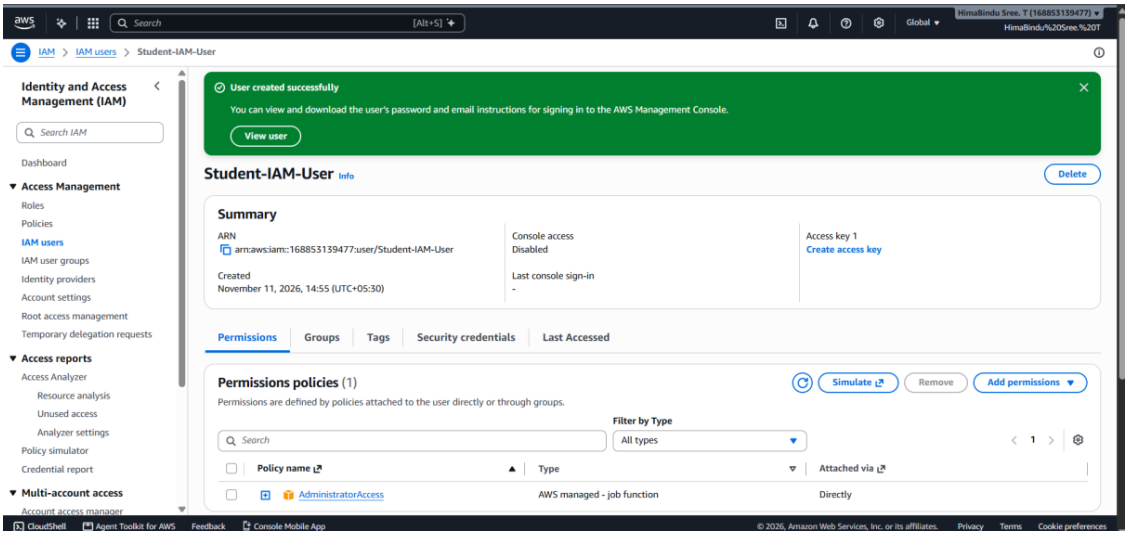


Figure 5: New IAM user page (Console access: Disabled)

Part B: Generating and downloading the access keys

Step 1: Open Security credentials

In IAM I clicked my new user and opened the Security credentials tab. I scrolled down to the Access keys section.

Step 2: Create the access key

I clicked Create access key and selected Command Line Interface (CLI) as the use case, because I would use the key with the AWS CLI. I ticked the confirmation box that acknowledges the recommendation, and clicked Next.

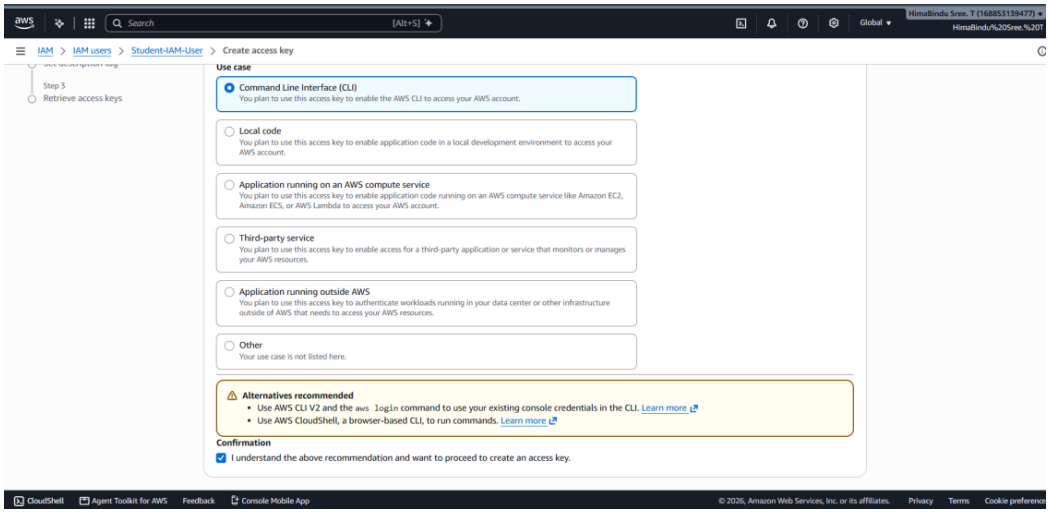


Figure 6: Access key use case – CLI selected and confirmation ticked

Step 3: Add a description

I entered a description tag (Student-CLI-Access-Key) and clicked Create access key.

Step 4: Download the credentials

AWS then showed the Access Key ID and the Secret Access Key. The secret key is shown only once, so I clicked Download .csv file straight away. I opened the file and checked that it has both the Access Key ID and the Secret Access Key, and then clicked Done. I keep this file private and do not share it with anyone.

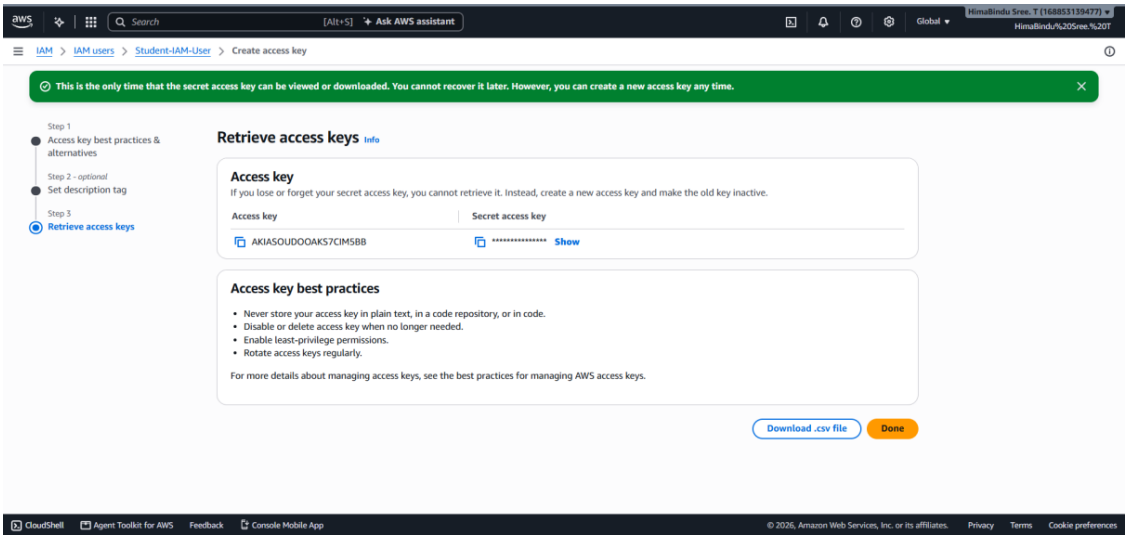


Figure 7: Retrieve access keys page

Part C: Turning on AWS Free Tier usage alerts

Step 1: Open Billing preferences

As the root user I searched for Billing and Cost Management and opened it. In the left panel I went to Preferences and clicked Billing preferences.

Step 2: Edit the alert preferences

In the Alert preferences section I clicked Edit and ticked Receive AWS Free Tier usage alerts. I typed my email address in the additional email box so that the alerts reach me.

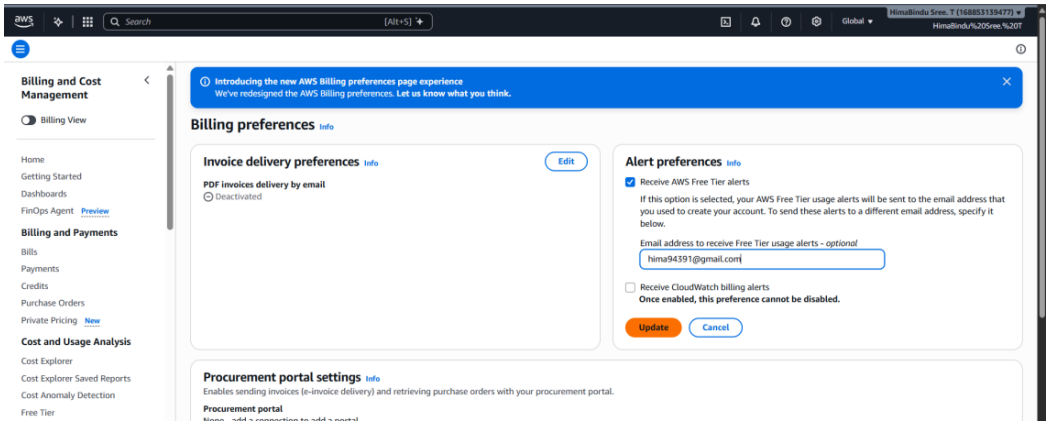


Figure 8: Alert preferences – Free Tier usage alerts ticked

Step 3: Save and verify

I clicked Update. AWS showed the message that my alert preferences were updated successfully, and the Alert preferences section now shows the AWS Free Tier alerts as delivered to my email address.

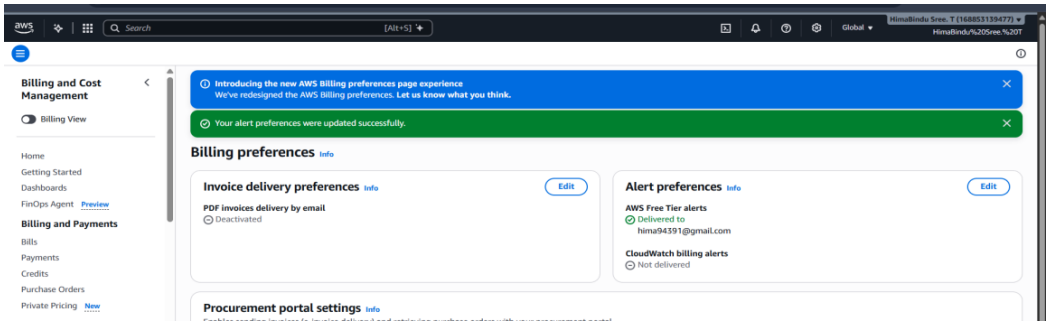


Figure 9: Alert preferences updated successfully

Part D: Setting up a billing alert with a budget

Step 1: Open Budgets

As the root user I opened Billing and Cost Management. Under Budgets and Planning I clicked Budgets and then clicked Create budget.

Step 2: Choose the template

Under Budget setup I kept Use a template (simplified) selected, and chose the Zero spend budget template, which sends a notification as soon as my spending goes above $0.01.

Step 3: Enter the name and email

I typed the budget name (AWS-Monthly-Cost-Budget) and my email address as the email recipient. The scope is all AWS services.

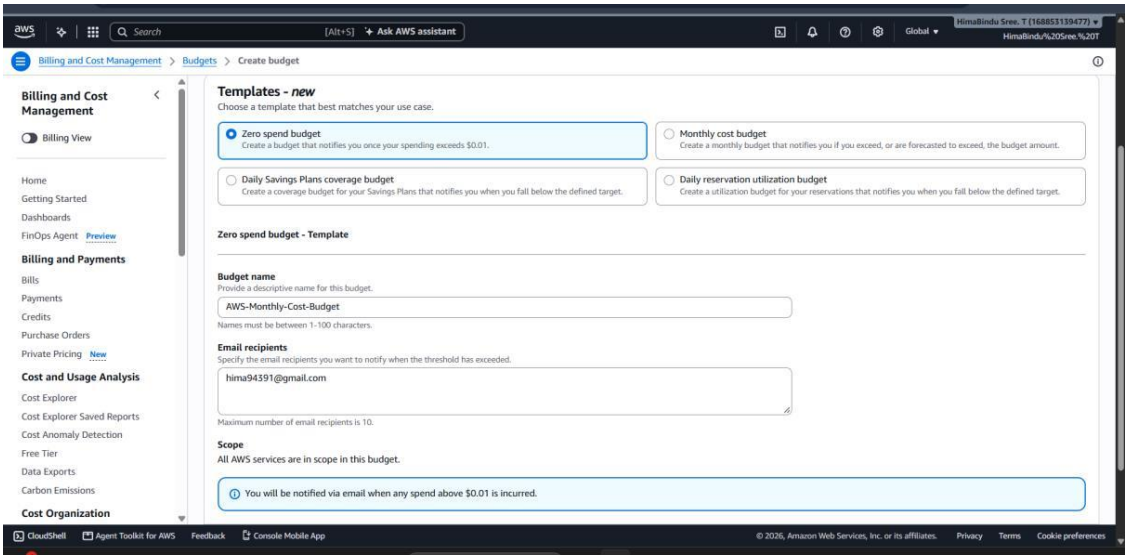


Figure 10: Zero spend budget template with budget name and email

Step 4: Create the budget

I clicked Create budget at the bottom of the page.

Step 5: Verify the budget

The new budget appears in the AWS Budgets dashboard with status OK and Healthy.

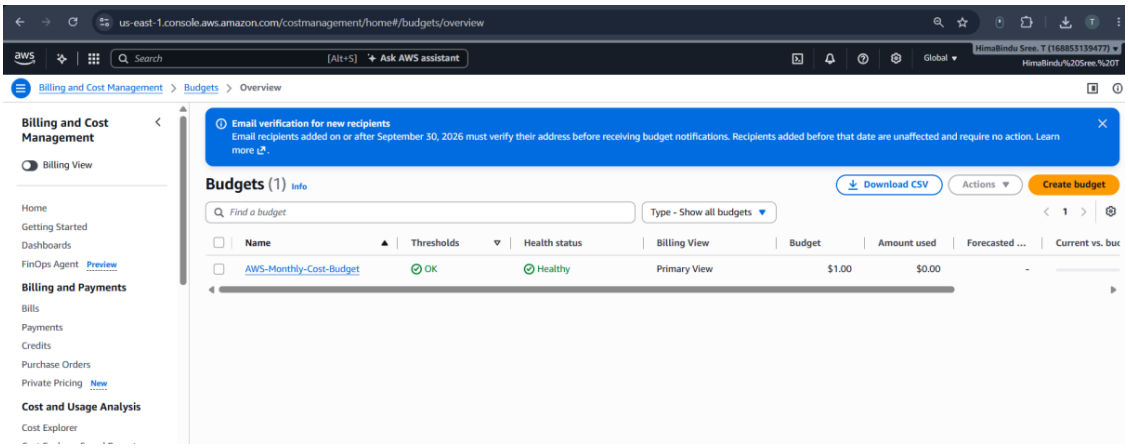


Figure 11: Budgets dashboard showing the new budget

Step 6: Check the alert

On the budget's Alerts tab, the alert Actual cost > $0.01 is set up and its threshold is Not exceeded. AWS sends a verification email to the recipient address, and that email has to be confirmed for the budget alerts to be delivered.

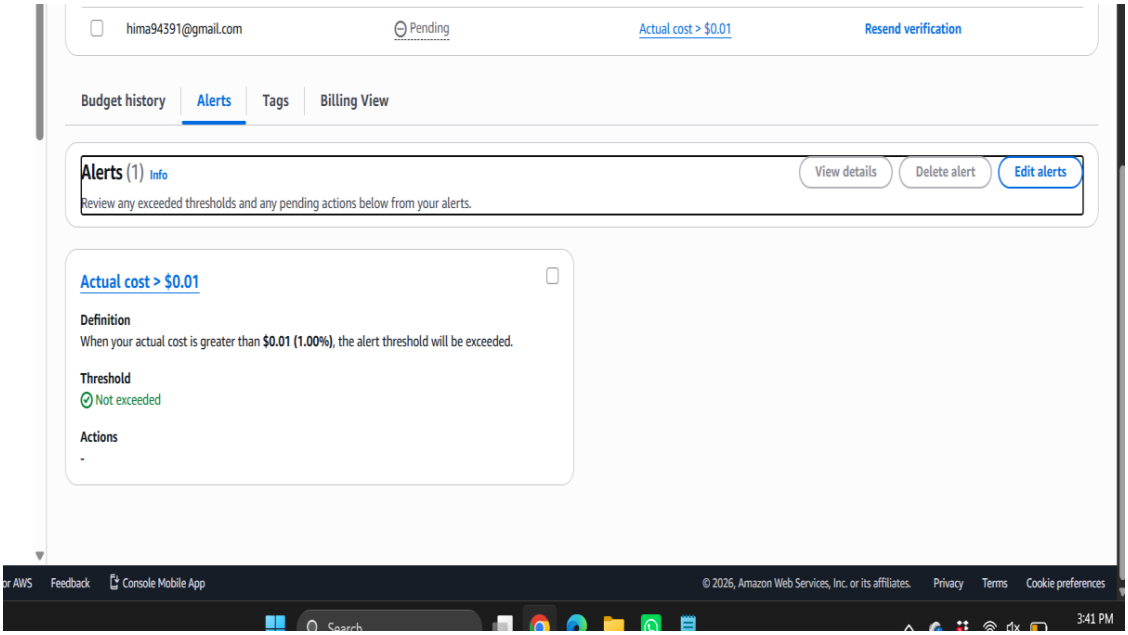


Figure 12: Budget alert – Actual cost > $0.01

Result

- An IAM user without console access was created successfully.
- The Access Key ID and Secret Access Key were generated and downloaded as a .csv file.
- AWS Free Tier usage alerts were enabled.
- A zero spend budget with an email alert was created, so I will be notified if my spending goes above $0.01.
- I understood that access keys must be stored safely and never shared.